

SWAN RIVER, MB, Canada

WMO: 714430

Lat: 52.1151N

Lon: 101.2330W

Elev: 1098

StdP: 14.12

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -27.0	(c) -22.8	(d) -37.5	(e) 0.7	(f) -24.5	(g) -32.6	(h) 0.9	(i) -22.3	(j) 21.5	(k) 8.6	(l) 19.7	(m) 9.3	(n) 6.7	(o) 220	(p) 0.658

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 21.6	(c) 84.9	(d) 67.9	(e) 81.5	(f) 65.6	(g) 78.6	(h) 64.0	(i) 70.6	(j) 81.0	(k) 68.3	(l) 77.9	(m) 66.3	(n) 75.4	(o) 9.2	(p) 220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
67.0	103.4	76.2	64.7	95.4	73.2	62.7	88.9	71.2	35.3	81.5	33.3	78.0	31.6	75.6	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
Min		Max		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
20.5	18.2	16.2		DB	-33.5	90.7	3.5	2.6	-36.0	92.6	-38.0	94.1	-40.0	95.5	-42.6	97.4
				WB	-33.8	75.1	3.6	2.9	-36.3	77.2	-38.4	78.9	-40.4	80.5	-43.0	82.6

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	35.2	1.8	5.4	18.2	36.5	49.9	60.0	65.0	62.7	53.6	39.5	21.4	6.9
	DBStd	25.03	15.22	13.79	13.84	11.17	8.91	6.71	5.44	6.06	7.69	8.37	12.19	14.30
	HDD50	6852	1493	1249	986	419	115	6	0	1	49	339	859	1336
	HDD65	11013	1958	1669	1451	855	471	176	68	116	349	790	1309	1801
	CDD50	1460	0	0	0	13	112	305	466	394	156	14	0	0
	CDD65	151	0	0	0	0	4	26	69	45	7	0	0	0
	CDH74	1937	0	0	0	6	123	329	737	579	158	5	0	0
	CDH80	450	0	0	0	1	25	75	167	147	35	0	0	0
Wind	WSAvg	7.3	7.7	7.6	7.4	7.9	8.4	7.2	6.4	6.3	7.0	7.2	7.4	7.5
Precipitation	PrecAvg	21.3	0.7	0.5	0.7	1.1	2.2	3.5	3.5	2.7	2.0	1.6	0.9	0.8
	PrecMax	25.7	1.7	1.5	2.5	2.6	5.2	6.9	8.4	7.2	4.3	4.2	2.9	1.5
	PrecMin	13.7	0.0	0.0	0.1	0.2	0.3	0.8	0.9	0.9	0.2	0.4	0.2	0.2
	PrecStd	3.2	0.5	0.4	0.6	0.6	1.2	1.6	1.8	1.4	1.1	0.8	0.6	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.5	38.0	50.2	73.1	82.8	87.1	88.5	89.1	84.4	71.3	52.6	39.6
		MCWB	34.6	34.1	42.2	53.3	59.5	67.3	71.3	69.7	63.3	56.1	43.6	34.6
	2%	DB	34.2	33.8	43.6	65.6	77.3	81.7	84.4	84.1	78.6	63.6	45.9	34.5
		MCWB	31.1	31.3	38.0	49.4	56.6	65.4	69.5	68.4	61.7	50.8	39.6	31.1
	5%	DB	28.9	30.4	39.7	59.9	72.2	77.7	81.2	80.2	73.5	57.8	40.3	29.7
		MCWB	26.7	28.3	35.5	47.0	54.3	62.3	67.1	65.7	59.1	47.7	35.8	27.3
	10%	DB	24.1	25.9	36.3	54.1	67.9	74.1	78.2	76.7	68.4	52.7	36.5	26.1
		MCWB	22.5	23.9	33.2	43.7	52.4	60.5	65.7	63.5	56.9	45.3	33.4	24.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.8	34.6	42.3	54.7	62.6	72.0	74.5	73.1	66.8	57.0	44.8	34.8
		MCDB	38.1	37.0	49.3	70.8	77.4	81.9	84.6	85.0	78.9	67.4	50.8	39.1
	2%	WB	31.4	31.3	38.0	50.8	58.7	67.4	71.1	69.9	63.5	52.2	39.9	31.2
		MCDB	34.0	33.2	42.7	63.5	72.6	77.1	80.9	80.9	74.5	61.0	45.5	34.2
	5%	WB	27.0	27.7	35.6	47.3	56.3	64.8	69.0	67.5	60.7	48.7	36.2	27.7
		MCDB	29.2	29.7	38.9	58.3	68.9	73.7	78.6	77.2	70.8	56.0	39.8	29.7
	10%	WB	22.6	23.5	33.3	44.4	53.7	62.5	67.0	65.4	57.9	46.1	33.6	24.5
		MCDB	24.2	25.3	35.9	53.5	65.1	71.4	75.7	74.2	66.5	52.0	36.4	26.3
Mean Daily Temperature Range	5% DB	MDBR	16.0	17.8	18.8	20.8	23.3	21.1	21.6	22.9	22.6	17.5	14.3	14.6
		MCDBR	17.6	17.7	18.9	30.7	31.7	27.5	26.6	29.0	32.1	27.8	19.3	17.8
		MCWBR	15.5	15.7	14.3	17.0	15.1	13.4	14.0	14.5	16.7	16.5	13.9	15.4
	5% WB	MCDBR	17.3	17.5	17.7	28.1	27.2	23.5	24.2	25.8	28.7	25.5	18.0	17.0
		MCWBR	15.4	15.6	13.7	16.5	14.0	12.8	14.0	14.1	15.8	15.6	13.3	14.9
Clear-Sky Solar Irradiance	taub		0.263	0.262	0.304	0.340	0.349	0.371	0.381	0.374	0.339	0.294	0.264	0.247
	taud		2.238	2.291	2.290	2.375	2.409	2.371	2.376	2.381	2.466	2.554	2.457	2.300
	Ebn at Noon		235	275	282	285	286	280	275	270	269	264	241	224
	Edh at Noon		21	26	33	34	35	36	36	34	28	21	17	16
All-Sky Solar Radiation	RadAvg		366	734	1207	1586	1768	1786	1893	1546	1118	635	324	258
	RadStd		27	46	74	122	164	140	89	133	111	85	46	28

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	+3.85	N/A	N/A	-0.99	N/A	N/A	N/A	N/A
Regional (10 neighbors)	N/A	N/A	N/A	N/A	-0.81	-1.03	N/A	N/A	N/A	N/A

Nomenclature: See separate page